MILITARY SPECIFICATION

ANODES, CORROSION PREVENTIVE, PLATINUM PLATED NIOBIUM WIRE TYPE, CATHODIC PROTECTION

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for prepared platinum plated niobium wire anodes and platinum plated niobium wire for impressed current cathodic protection application in the reserve fleet (inactive ships), and for temporary cathodic protection systems.

1.2 Classification. The anodes shall be of the following types, as specified (see 6.2).

Type I - Prepared platinum plated niobium wire anode.
Type II - Platinum plated niobium wire.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5340

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MIL-P-116 - Preservation, Methods of.
MIL-C-915 - Cable and Cord Electrical, for Shipboard Use General Specifications for.
MIL-I-17695 - Insulation Tape, Electrical, Filler Type, Flameproof, Synthetic.
MIL-I-81765 - Insulating Components, Molded, Electrical, Heat Shrinkable, General Specification for.
MIL-I-81765/5 - Insulating Components, Molded, Electrical, Heat Shrinkable Butyl Rubber, Flexible, Crosslinked.

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Construction. The prepared platinum plated niobium anodes (type I) shall be constructed of 5 feet ± 3 inches of platinum plated niobium wire, coupled to a 30-foot length of electrical cable. The platinum plated niobium wire (type II) shall be a continuous 100 foot coil.

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3.2 Material.

3.2.1 Platinum plating. The platinum plating shall consist of 99.9 percent pure platinum on an etched surface to an average thickness of 100 micro inches (0.0001 inch).

3.2.2 Niobium wire. The wire shall be 99.9 percent pure, annealed niobium wire, with a diameter of 0.093 ± 0.003 inches.

3.2.3 Electrical cable. The cable shall be type SHOF 23 conforming to MIL-C-915.

3.2.4 Watertight connection (type I). The wire to cable connection shall be watertight by use of insulation (putty) 3/16 ± 1/32 inch thick conforming to MIL-I-17695 and a protective top covering of a water resistant, heat shrinkable tubing conforming to MIL-I-81765 and MIL-I-81765/5 (see 4.3.3). The electrical connection between the electrical cable and the platinized niobium wire shall utilize a mechanical fastener or soldering technique or both. The electrical resistance of the connection shall be less than 0.01 ohm (see 4.3.3). The connection shall be able to support a 10-pound pull (see 4.3.3).

3.3 Workmanship.

3.3.1 Platinum plated niobium wire (types I and II). The surface of the platinized niobium wire shall be smooth and free of dents, kinks, and other defects of a similar nature that may affect serviceability, consistent with good commercial practice.

3.3.1.1 Adhesion. Adhesion of platinum plating to niobium wire shall be tested in accordance with 4.3.1.

3.3.2 Watertight connection (type I). The surface of the heat shrinkable tubing shall be free of splits, cracks, burn spots, and other defects of a similar nature that may affect serviceability, consistent with good commercial practice and in accordance with 4.3.3.

3.3.3 Electrical cable (type I). The surface of the electrical cable shall be free of kinks, dents, cracks and other defects of a similar nature that may affect serviceability, consistent with good commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection.

4.2.1 Lot. For the purpose of sampling, a lot shall consist of all anodes of the same type from the same melt or production run offered for delivery at one time. However, this lot shall not exceed a maximum of 200 prepared anodes or 1,000 feet (ten 100 foot coils) of platinized-niobium wire.

4.2.2 Sampling for examination (type I). A random sample of prepared anodes shall be selected from each lot in accordance with MIL-STD-105 at inspection level II. In terms of defects per 100 units, the acceptable quality level (AQL) shall be 4.0 for total defects. Each sample specimen shall be examined to assure compliance with 3.3.1, 3.3.2, and 3.3.3.

4.2.3 Sampling for examination (type II). The sample for type II shall be a length of wire selected at random from the lot. The length of the sample, in feet, shall be determined in accordance with MIL-STD-105 at inspection level I. In terms of defects per 100 feet, AQL shall be 4.0 for total defects.

4.2.4 Sampling for testing. Sampling for the purpose of testing shall be conducted as dictated by the individual test procedures of 4.3.

4.2.5 Noncompliance. If a sample or lot fails to pass any of the quality assurance provisions of section 4, it shall be considered rejected. Rejected items shall be returned to the contractor for replacement. Shipping of rejected items shall occur at no expense to the Government.

4.3 Test procedures.

4.3.1 Adhesion. Sampling for adhesion tests shall be one type I anode per hundred. If the sample specimen fails, three randomly chosen anodes shall be tested. If two out of three sample specimens fail, the lot shall be rejected. Type II anode sampling for adhesion test shall be performed on one end of one coil. If the sample specimen fails, tests shall be run at three randomly chosen locations along the length of the wire with a minimum separation of 10 feet for each test. If two out of three sample specimens fail, the lot shall be rejected. Adhesion may be determined by scraping the surface of the plated wire to expose the base metal and examining at four times (x) magnification for evidence of nonadhesion. Alternatively, the plated wire may be clamped in a vise and the projecting portion bent back and forth until rupture occurs. Adhesion shall not be satisfactory if the edge of the ruptured plate can be peeled back or if separation between the plate and the base metal can be seen at the point of scraping or rupture when examined at four times (x) magnification.

4.3.2 Plating thickness. The platinum plate thickness shall be tested in accordance with FED-STD-151, method 520.1 at two separated points of each type I or II anode using the sampling and rejection or 4.3.1. The average of the two separated points shall be considered a single sample. The plating thickness shall comply with 3.2.1.

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4.3.3 Watertight connection (type I). Sampling and rejection for watertight connection tests shall be in accordance with that of adhesion tests for type I anodes (see 4.3.1). The sample specimens used for destructive adhesion tests (see 4.3.1) may be used as specimens for the watertight connection test. Compliance with the requirements of 3.2.4 shall be determined by the following test.

(a) Pull test. The sample specimen shall have the wire to cable connection tested for a 10-pound tension load, for 48 continuous hours.
(b) Electrical resistance. The cable of the sample specimen shall be cut 6 inches above the anode connection. The resistance between the cable and the wire shall be measured using a meter with a resolution better than 0.1 milliohm. The resistance shall be less than 0.01 ohm.
(c) Connection insulation. The sample specimen shall be severed at the connection and examined. The thickness of the insulation tape (putty) shall be 3/16 ± 1/32 inch.

4.3.4 Wire thickness. Each type II anode coil shall be tested for wire thickness. The sample specimen shall have each coil end attached to an ohm meter with a resolution better than 0.1 milliohm. The specimen shall register a maximum resistance of 1.69 ohms per 100 feet at temperature 20 ± 4°C. If the resistance is greater than 1.69 ohms per 100 feet, the specimen shall be declared unsatisfactory.

4.4 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Anodes shall be preserved-packaged in accordance with method III of MIL-P-116 and as follows:

(a) Type I anodes. Each prepared anode shall be fully extended with the 30-foot lead cable coiled and shall be individually protected by wrapping, interleaving or by use of a easily removable covering such as extruded polyethylene flexible tubing. Ten type I anodes shall be bundled and packed as specified in 5.2.
(b) Type II anodes. Each wire shall be furnished in 100-foot coil with a minimum diameter of 1-1/2 feet. The wire shall be protected as specified in 5.1.1(a). Each coil shall be packaged in a corrugated fiberboard box and packed as specified in 5.2.

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5.1.2 Level C. Anodes shall be individually packaged to afford protection against corrosion, deterioration, and physical damage during a shipment from the supply source to the first receiving activity for immediate use. The contractor's normal retail or wholesale preservation-packaging methods may be utilized when such meet the requirements of this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Anodes packaged as specified (see 6.2), shall be packed in class weather-resistant fiberboard boxes conforming to PPP-B-636, grade V15c, with type, variety and style at the contractor's option. Box closure, waterproofing and reinforcement shall conform to method V of the appendix to the box specification.

5.2.2 Level B. Anodes, packaged as specified (see 6.2), shall be packed in class domestic fiberboard boxes conforming to PPP-B-636 minimum grade 275 with type, variety and style at the contractor's option. Box closure shall conform to method I in accordance with the appendix of the box specification.

5.2.3 Level C. Anodes packaged as specified (see 6.2), shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing or method of shipment shall comply with Uniform Freight or National Motor Freight Rules or Regulations or other carrier rules as applicable to the mode of transportation.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Anodes covered by this specification are intended for use with impressed current systems for cathodic protection of underwater surfaces of inactive ships. They are primarily designed for suspension in the water alongside the ship's hull in inactive fleet locations. Anodes are also intended for use with temporary cathodic protection systems.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Types of anode required (see 1.2).
(c) Levels of preservation-packaging and packing required (see 5.1, 5.2, 5.2.1, 5.2.2, and 5.2.3.)

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
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(Project 5340-N084)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-24625(SH)	ANODES, CORROSION PREVENTIVE, PLATINUM PLATED NIOBIUM WIRE TYPE, CATHODIC PROTECTION

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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